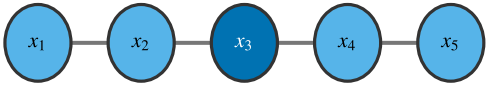
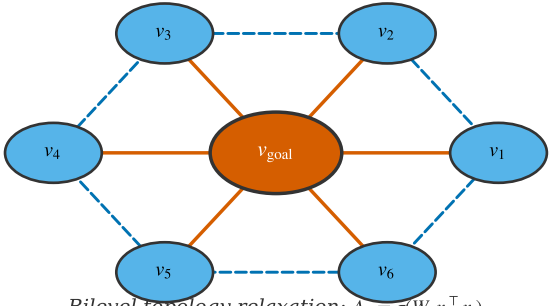
**(a) Traditional 1D Lattice: Diameter $\Omega(b)$**




*Fixed nearest-neighbor kernel* $W_L, W_C, W_R$

Receptive field expands as $O(T)$ steps

**(b) Meta-Learned Graph NCA: Diameter** $\leq 2$ **(**$O(1)$ **Receptive Field)**




*Bilevel topology relaxation:* $A_{ij} = \sigma(W_\theta x_i^\top x_j)$